



The economic implications of a network SIR-Macro model of epidemics

Keven R.A. André^a, Marcelo Arbex^{b,*}, Márcio V. Corrêa^{a,2}

^a CAEN - Graduate Studies in Economics, Federal University of Ceará, Brazil

^b Department of Economics, University of Windsor, Canada

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ABSTRACT

We embed social networks in a conventional SIR-Macro framework. The network effect amplifies the negative effects of an epidemic. Greater connectedness at work has stronger recessionary effects. The network average degree can be interpreted as policies such as social distancing.

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1. Introduction

We reconsider a conventional SIR-Macro framework (Eichenbaum et al., 2021; Kermack and McKendrick, 1927) to embed a social network model along the lines of models of the transmission of information in large, complex networks (Calvó-Armengol and Jackson, 2007; Arbex et al., 2019). A Network-SIR-Macro model can accommodate differences in the number of interactions between individuals in a population. The probability of becoming infected depends not only on the individual's consumption and work decisions (Eichenbaum et al., 2021) but also on the number of people she interacts with during these activities. Departing from the standard SIR model, individuals interact mostly within narrower groups (e.g. workplaces and social environments) - a key insight is to account for the heterogeneity of contacts between susceptible and infected individuals when they engage in consumption and work activities. Individuals are more likely to become infected if their number of interactions is higher. A greater connectedness of those who get infected implies that they are also more likely to pass on the disease themselves and contributes to accelerating the spread of the disease in the early stages.

The epidemic induces a recession (Eichenbaum et al., 2021). However, the network effect amplifies the negative effects of an epidemic - the recession is more severe and occurs sooner in economies where individuals interact with more contacts while working or consuming. We consider heterogeneity regarding number of contacts in consumption and work and we find that greater connectedness at work (*vis-à-vis* at consumption) has a stronger recessionary effect - if individuals interact (on average) with more contacts at work, the drop in aggregate consumption is larger.

The economic literature on the COVID-19 epidemic has developed at a remarkable pace.³ One of the major assumptions of the classic SIR model is uniform mixing across the entire population - infections evolve as if any susceptible individual interacts with and could be infected by any infected individual across the population with equal probability (Eichenbaum et al., 2021; Craig et al., 2020). However, it is well documented that social interactions are not organized in this stylized way (Calvó-Armengol and Jackson, 2007; Jackson, 2008; Banerjee et al., 2013; Elliott et al., 2014) - individuals interact mostly within much narrower groups shaped, e.g. by work and social environments. In our model, the probability of becoming infected and infection rates depend on the number of people agents interact with in market (consumption- and work-related) activities.⁴ In the face

* Corresponding author.

E-mail addresses: arbex@uwindsor.ca (M. Arbex), marciocorreia@caen.ufc.br (M.V. Corrêa).

¹ CAEN/UFC International Visiting Professor.

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³ See Ellison (2020) for a survey of results on the classic SIR model and variants allowing for heterogeneity in contact rates. See also, Alfaro et al. (2020), Bethune and Korinek (2020), Berger et al. (2020), Acemoglu et al. (2021), Brodeur et al. (2021).

⁴ The model can be extended to include infections generated by non-market activities as in Eichenbaum et al. (2021).

of a pandemic, a susceptible individual interacting with many infected contacts will be more likely to become infected than an individual with only few or no infected contacts in their network. Reductions in the average degree in the network (disease transmission technology) can be interpreted as policies such as social distancing measures and mandatory face masks, leading to less interaction among agents in the economy and thereby slowing down the epidemic.⁵

2. A network SIR-macro model

2.1. The outbreak of an epidemic

The population is divided into susceptible (S), infected (I), recovered (R), and deceased individuals (Eichenbaum et al., 2021); m_t^S , m_t^I , m_t^R , and m_t^D , are their fractions in the population, respectively. The outbreak of an epidemic occurs when a fraction ε of susceptible individuals is infected by a virus (i.e., $m_0^I = \varepsilon$, $m_0^S = 1 - \varepsilon$).⁶

2.2. Network structure and infection transmission

We assume that agents are in contact to one another in a network, whose structure is exogenous (Arbex et al., 2019). A network is described by a degree distribution $\{D_z\}_{z=1}^{\infty}$, where D_z is the proportion of agents who are in contact with $z \in [1, \infty)$ agents. We assume power-law distributions (agents that interact with many agents are more likely to be infected) and apply the mean field approach (Vega-Redondo, 2007). In our model, contacts are meant to represent the number of people an individual interacts with via consumption or work (e.g., people that do (eat) groceries at the same supermarket (restaurant), or work at the same firm). Each agent may have contacts to whom she passes the virus when infected, and from whom she may receive the virus when susceptible.

The probability a given individual interacts with other s individuals, either via consumption or via work, is $\psi_s = (sD_s) / \langle z \rangle$, where $\langle z \rangle = \int_{z=1}^{\infty} (zD_z) dz$ is the average degree in the network.⁷ Note that $\psi_s \neq D_s$, i.e., the probability one of the individuals you interact with s individuals, is not equal to the proportion of the population that interacts with s individuals. This gives the probability that an individual that interacts with s individuals passes another susceptible individual the virus.

The infection rate among agents that interact with s individuals is $m_{s,t}^I$. These agents contact susceptible agents either via consumption ($x = c$) or via work ($x = n$) with probability ρ_t^x . Infected agents (I) pass the virus onto other susceptible peers (S) with probability $\varphi(x_t^I, x_t^S) = (m_t^S x_t^S + m_t^I x_t^I)^{1-\lambda_x}$, where λ_x measures the efficacy of this technology. When λ_x is close to 0 (1), the virus is transmitted at a higher rate and requires less interaction among individuals (transmission is more difficult and will depend more on individuals' consumption or work). We assume $\langle z \rangle = \langle z \rangle_c = \langle z \rangle_n$ (benchmark). The average degree in the network $\langle z \rangle$ (disease transmission technology λ_x) can be interpreted as policies such as social distancing measures.

For a given agent and activity $x = \{c, n\}$, the probability another agent of type s is infected, meet a susceptible individual,

transmits the virus, and is a peer is $m_{s,t}^I \rho_t^x \varphi(x_t^I, x_t^S) \psi_s$. Integrating over all possible s , the probability a susceptible agent is infected by a peer via an activity x is

$$\begin{aligned} \Omega_t^x &= \int_{s=1}^{\infty} m_{s,t}^I \rho_t^x \varphi(x_t^I, x_t^S) \psi_s ds \\ &= \rho_t^x \varphi(x_t^I, x_t^S) \frac{1}{\langle z \rangle} \int_{s=1}^{\infty} m_{s,t}^I (sD_s) ds \\ &= m_t^I \rho_t^x \varphi(x_t^I, x_t^S). \end{aligned} \quad (1)$$

The probability a susceptible agent of type z receives the virus from at least one peer is $p_t^x = 1 - (1 - \Omega_t^x)^z$, $x = \{c, n\}$, and the aggregate probability susceptible agents of different types z receive the virus is

$$P_t^x = \int_{z=1}^{\infty} p_t^x D_z dz. \quad (2)$$

The timing convention is as in Eichenbaum et al. (2021). At the end of the period, the number of newly infected individuals T_t is

$$T_t = (m_t^S P_t^c)^\gamma (m_t^S P_t^n)^{(1-\gamma)} \quad (3)$$

where γ is the relative weight of infection via consumption, $m_t^S P_t^c$ ($m_t^S P_t^n$) represents the number of newly infected people through consumption (work).

The laws of motion of the fraction of susceptible, infected, recovered and number of deaths in the population are, respectively

$$m_{t+1}^S = m_t^S - m_t^S (P_t^c)^\gamma (P_t^n)^{(1-\gamma)} = m_t^S \left(1 - (P_t^c)^\gamma (P_t^n)^{(1-\gamma)}\right). \quad (4)$$

$$m_{t+1}^I = (1 - \pi_R - \pi_D) m_t^I + T_t, \quad (5)$$

$$m_{t+1}^R = m_t^R + \pi_R m_t^I \quad (6)$$

$$m_{t+1}^D = m_t^D + \pi_D m_t^I, \quad (7)$$

where π_R represents the rate at which infected people recover from the disease and π_D is the mortality rate. Total population in period $t + 1$ is $Pop_{t+1} = Pop_t - \pi_D m_t^I$; with $Pop_0 = 1$.

2.3. Susceptible, infected, and recovered individuals' problems

The instantaneous utility of a type- J , $J = \{S, I, R\}$, agent is

$$u(c_t^J, n_t^J) = \ln c_t^J - \frac{\theta}{2} (n_t^J)^2. \quad (8)$$

A susceptible agent's problem is to maximize lifetime utility

$$U_t^S = u(c_t^S, n_t^S) + \beta[(1 - \tau_t) U_{t+1}^S + \tau_t U_{t+1}^I], \quad (9)$$

subject to the budget constraint:

$$(1 + \mu_{ct}) c_t^J = w_t^J n_t^J + \Gamma_t^J, \quad (10)$$

The probability τ_t a susceptible agent becomes infected is

$$\tau_t = \frac{T_t}{m_t^S} = (P_t^c)^\gamma (P_t^n)^{(1-\gamma)}. \quad (11)$$

A susceptible agent's problem is to maximize lifetime utility, Eq. (9), subject to constraints (10) and (11). The first-order conditions with respect to consumption, hours worked and τ_t are, respectively,

$$u_1(c_t^S, n_t^S) - \eta_{bt}^S (1 + \mu_{ct}) + \eta_{rt} \gamma \left(\frac{P_t^n}{P_t^c}\right)^{1-\gamma} \frac{\partial P_t^c}{\partial c_t^S} = 0, \quad (12)$$

$$u_2(c_t^S, n_t^S) + \eta_{bt}^S w_t^S (1 - \mu_{nt}) + \eta_{rt} (1 - \gamma) \left(\frac{P_t^c}{P_t^n}\right)^\gamma \frac{\partial P_t^n}{\partial n_t^S} = 0, \quad (13)$$

$$\beta(U_{t+1}^I - U_{t+1}^S) - \eta_{rt} = 0, \quad (14)$$

⁵ Vaccines and lockdowns are beyond the scope of this paper. Vaccines affect the spread of the epidemic through channels beyond disease transmission. Lockdowns would imply the necessity of interactions effects in the production process with potentially countervailing network effects. We leave this for future research.

⁶ Preinfection economy follows (Eichenbaum et al., 2021).

⁷ The degree of a node in a network is the number of links connecting it to other nodes. Average degree $\langle z \rangle$ is the mean of all the node degrees in a network (Jackson, 2008).

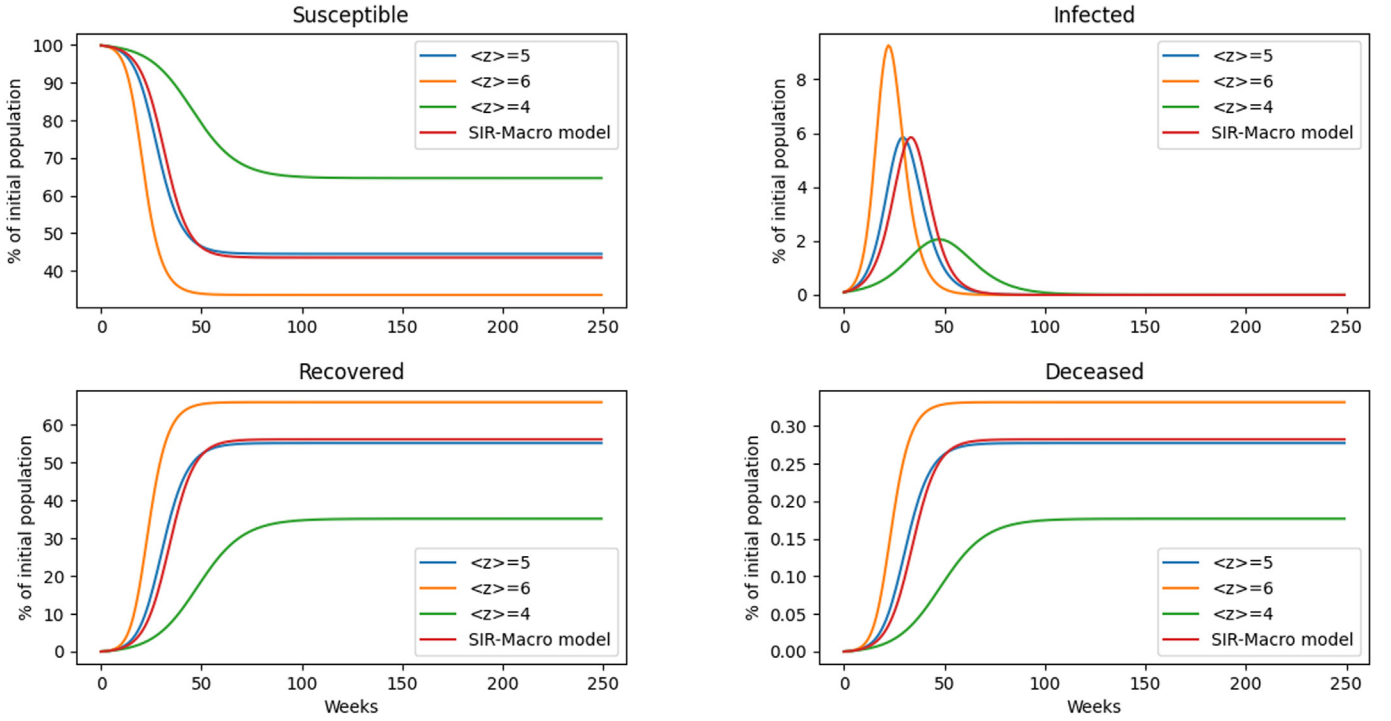


Fig. 1. Network SIR model – progression of an epidemic.

where η_{bt}^S and η_{rt} are the Lagrange multipliers associated with Eqs. (10) and (11), respectively. The last terms of Eqs. (12) and (13) capture the mechanism through which a susceptible agent internalizes the risk of getting infected when she interacts with certain number of people in consumption and work activities.

The lifetime utility of an infected agent (U_t^I) and a recovered individual (U_t^R) are, respectively,

$$U_t^I = u(c_t^I, n_t^I) + \beta [(1 - \pi_R - \pi_D) U_{t+1}^I + \beta \pi_R U_{t+1}^R], \quad (15)$$

$$U_t^R = u(c_t^R, n_t^R) + \beta U_{t+1}^R. \quad (16)$$

There is a continuum of competitive representative firms of unit measure that produce a single good (Y_t) using labor only, i.e., $Y_t = A \left(\sum_J \phi^J N_t^J \right)$, where $N_t^J = m_t^J n_t^J$ and ϕ^J denotes the labor productivity of an individual $J = \{S, I, R\}$. Firm's profit maximization $\Pi_t = A \left(\sum_J \phi^J N_t^J \right) - \sum_J w_t^J N_t^J$ implies $A\phi^J = w_t^J$, for $J = \{S, I, R\}$ in every period t .

The government budget constraint is: $\mu_{ct} \left(\sum_J m_t^J c_t^J \right) + \mu_{nt} \left(\sum_J w_t^J N_t^J \right) = \Gamma_t \left(\sum_J m_t^J \right)$. In equilibrium, each agent solves her maximization problem and the government constraint is satisfied in each period. The goods market and labor market clear: $\sum_J m_t^J c_t^J = AN_t$, $\sum_J m_t^J n_t^J = N_t$, and $c_t^J = C_t^J$, $n_t^J = N_t^J$ for $J = \{S, I, R\}$ and every t .

3. Quantitative analysis

Our calibration strategy is to choose parameters such that the share of the initial population that is infected in the Network SIR-Macro model peaks at a level similar to Eichenbaum et al. (2021)'s model - 5.84% (5.8%); week 29 (33), respectively (Fig. 1, top-right panel). In the context of our Network-SIR-Macro model, there are no available estimates in the literature regarding parameters related to the network structure. We then follow previous works regarding the average number of contacts ($\langle z \rangle$) and the power-law exponent a . This strategy allows us to adjust the probability ρ_t^x agents contact susceptible agents either via consumption ($x = c$)

or via work ($x = n$) to match Eichenbaum et al. (2021)'s health outcomes.⁸

Table 1 presents our baseline parameters. The degree distribution is a power-law distribution $D_z = (a - 1)z^{-a}$. We assume agents are, on average, in contact with five individuals, $\langle z \rangle = 5$, with $a = 2.25$ (Arbex and O'Dea, 2014; Arbex et al., 2016). The transmission of the virus from infected agents (I) to their susceptible peers (S) is commanded by $\lambda_c = \lambda_x = 0.95$ (Arbex et al., 2016). Virus transmission via consumption and labor are equally important ($\gamma = 0.5$) to the number of newly infected individuals (T_t). Finally, we set $\rho_t^x = 0.085$, $x = c, n$.⁹

Fig. 1 illustrates the impact of heterogeneity in the number of contacts on the progression of an epidemic. Compared to the benchmark case ($\langle z \rangle = 5$) (Eichenbaum et al. (2021)'s baseline SIR model), if agents have more contacts on average (i.e., $\langle z \rangle = 6$) the disease initially spreads significantly faster through more connected individuals, but then the spread slows down and the number of infected approaches zero earlier. This reflects the fact that as the infection progresses, fewer individuals remain susceptible and are infected, and the share of recovered individuals increase.

The epidemic induces a recession (Eichenbaum et al., 2021): aggregate consumption and hours worked fall. However, the recession is more severe and occurs sooner in economies where individuals interact with more contacts in consumption and work activities (e.g., $\langle z \rangle = 6$; Fig. 2). The network effect amplifies the negative effects of an epidemic. The virus causes more people to be infected earlier in an epidemic and, since infected people are less productive ($\phi^I = 0.80$), they work less hours, earn relatively less income, and, consequently, consume less. Because the share of deceased individuals is higher in more connected economies,

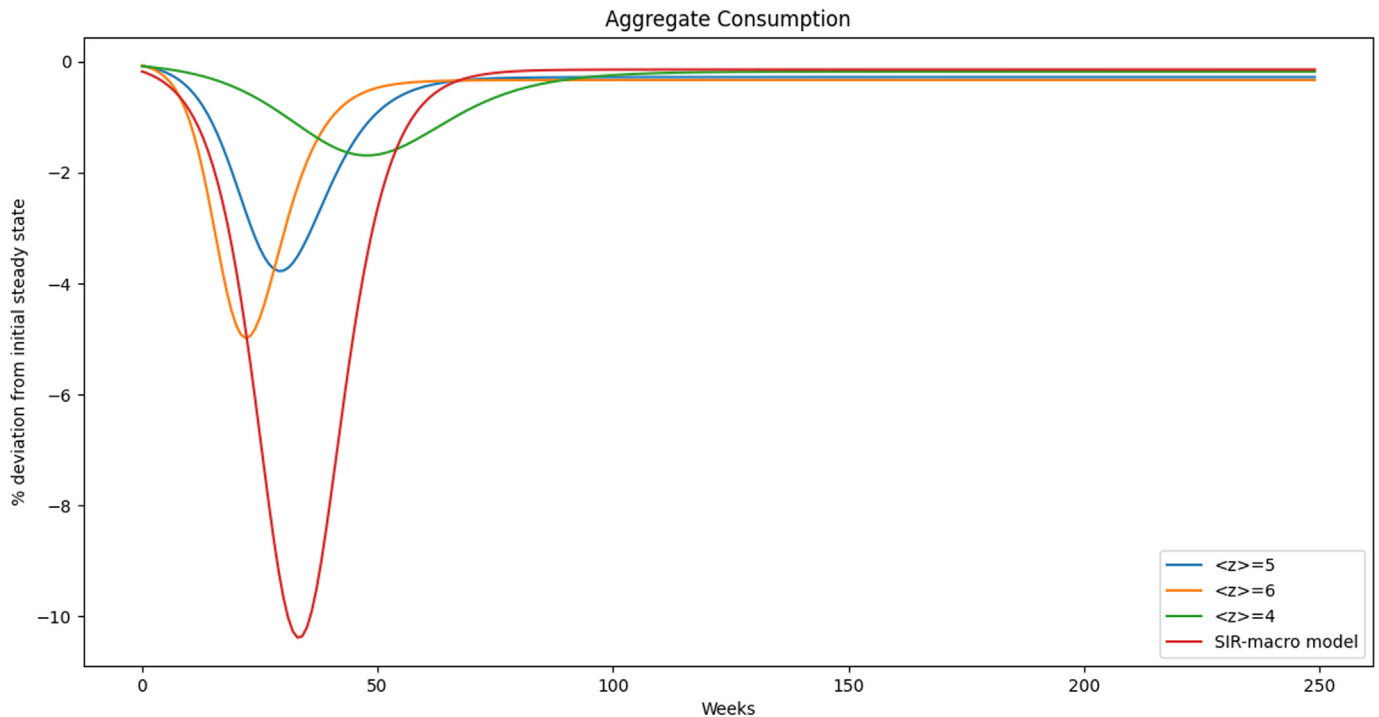
⁸ A calibration strategy that targets Eichenbaum et al. (2021)'s health and economic outcomes would involve recalibrating $\langle z \rangle$ along with other parameters of the model - a reinterpretation of Eichenbaum et al. (2021) parameters π_1 , π_2 , and π_3 .

⁹ We conduct robustness analysis for several alternative parameter configurations (results available upon request).

Table 1
Model parameters.

Preferences, technology				SIR model			Network component				
(1)	(1)	(1)	(1)	(1)	(1)	(1)	(2)	(2)	(2)	(3)	(4)
β	θ	A	ϕ^I	π_D	π_R	ε	a	$\langle z \rangle$	λ_x	γ	ρ^x
0.96 ⁵²	0.001275	39.835	0.80	$\frac{(7 \times 0.005)}{18}$	$\frac{7}{18} - \pi_D$	0.001	2.25	5	0.95	0.50	0.085

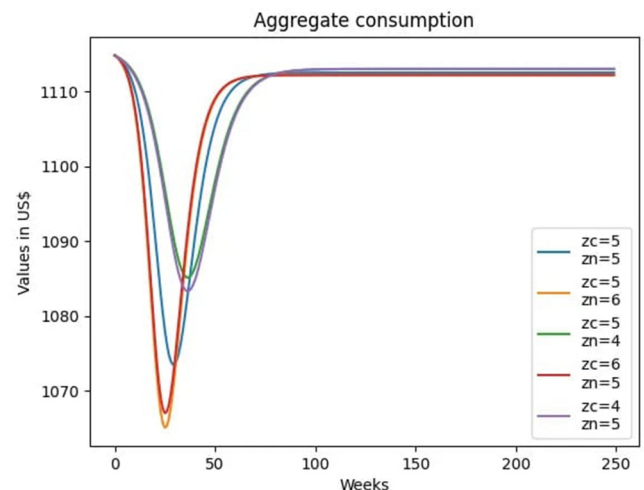
Note: $x = \{c, n\}$; (1) Eichenbaum et al. (2021); (2) Arbex et al. (2019, 2016); (3) Benchmark value; (4) Calibrated.

**Fig. 2.** Network SIR model: Aggregate consumption (C), $\langle z \rangle = \langle z_c \rangle = \langle z_n \rangle$.

the epidemic permanently reduces the size of the work force. This also contributes to the observed fall in aggregate consumption and hours worked in more connected economies.

Allocations - e.g., aggregate consumption (Fig. 2) - are not identical in the Basic SIR-Model and in a Network-SIR-Model (baseline: $\langle z \rangle = 5$). The reason is twofold: (i) we target the health results in our calibration exercise (Fig. 1); (ii) virus transmission dynamics is substantially different in the two models. In particular, the first two terms of Eqs. (12) and (13) are similar to those present in the Eichenbaum et al. (2021)'s equilibrium conditions. However, the last terms differ from those of a SIR-Macro model and they are a direct implication of a Network-SIR-Macro model and the network structure we consider, capturing the mechanism through which a susceptible agent internalizes the risk of getting infected when she interacts with certain number of people in consumption and work activities.¹⁰

Different combinations of $\langle z_c \rangle$ and $\langle z_n \rangle$ highlight the effects of heterogeneity in the number of (average) contacts in consumption and work for the progression of an epidemic (Figure A.1, Appendix A.1), as well as the aggregate measures of consumption and hours worked. Our results suggest that greater connectedness at work (*vis-à-vis* at consumption) has a stronger recessionary effect (Fig. 3). Relative to the benchmark $\langle z_c \rangle = \langle z_n \rangle = 5$, if individuals interact (on average) with more contacts at work

**Fig. 3.** Aggregate consumption, hours worked - $\langle z_c \rangle$, $\langle z_n \rangle$.

than during consumption, i.e., $\langle z_c \rangle = 5$; $\langle z_n \rangle = 6$ versus $\langle z_c \rangle = 6$; $\langle z_n \rangle = 5$, the drop in aggregate consumption is bigger – more people get infected at work, production falls, and, consequently, consumption falls.

A larger fraction of the population is infected faster the more efficiently the virus is transmitted, i.e., higher λ (Figure A.4, Appendix A.2). The lower the contact probability ($\rho^c = \rho^n$), the

¹⁰ In the context of a Network-SIR-Macro model and economic outcomes, the results presented in Fig. 2 could be interpreted as the case of a large $\langle z \rangle$; i.e., susceptible individuals are in contact with many individuals and can be infected with equal probability.

lower the share of the population that becomes infected by the virus and the longer it takes to happen (Figure A.5, Appendix A.2). As agents are contacted and infected at work with higher probability, they become sick and are less productive. Their labor income is lower, which lowers consumption and, consequently, the economy's output.

4. Conclusion

We embed social networks in a conventional SIR-Macro framework. Recession is more severe and occurs sooner in economies where individuals interact with more contacts while working or consuming. Significant heterogeneity in connections can accelerate the early spread of disease while reducing the share of the population that is affected in the long run – the network effect amplifies the negative effects of an epidemic. Greater connectedness at work has a stronger recessionary effect.

Data availability

We use public available data.

Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.econlet.2023.111047>.

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